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Introduction

10.1 Advanced Applications of Relative Strength

Introduction

In Level I, we introduced the concept of relative strength and revealed how it is calculated and used to identify securities outperforming a benchmark. We also introduced the idea of a relative strength ranking system that can be used to rank a basket of securities into a list from strongest to weakest. While the ranking system is a great way to incorporate relative strength into portfolio analysis, there is an enormous amount of processing involved.

In this unit, we introduce a modern way of classifying securities using relative strength, visualizing the relative performance of many securities on a single chart, and solving the biggest issues with relative strength analysis when comparing multiple securities.

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10.1 Advanced Applications of Relative Strength



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10.1 Advanced Applications of Relative Strength

The Challenges of Relative Strength Analysis

While the relative strength line can be helpful for the technical analyst, there are two major issues with relative strength to be aware of:

1. Relative trend can be opposite to absolute trend.
2. Numerical relative strength values are meaningless.

Relative Trend can Be Opposite Absolute Trend

This is an issue discussed in Level I. A security may be in an absolute downtrend, but if the benchmark is trending down more steeply, the relative strength line will be in an uptrend (and vice versa).

Figure 10.1.1 shows this issue; the security is in an uptrend, but the relative strength line (in red) is in a downtrend. In this situation, we can assume the benchmark must be performing better than the security, leading to a negative trend on the relative strength line.

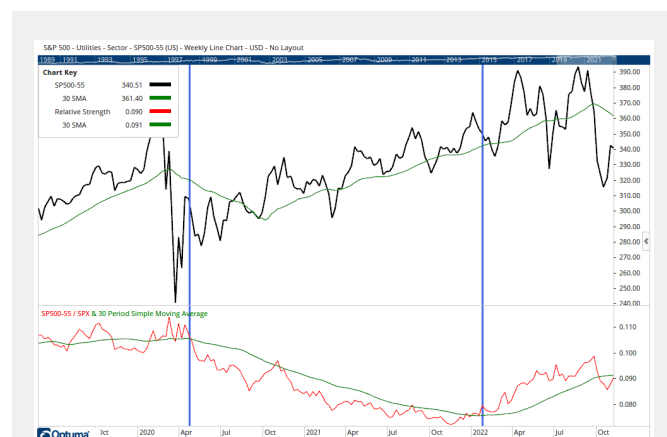


FIGURE 10.1.1 S&P 500 Utilities index and relative strength line vs. S&P 500, each with a 30-period moving average

Figure 10.1.1 shows the S&P 500® Utilities index and its relative strength line vs. the S&P 500 index. Thirty-week simple moving averages are overlaid on price and on the relative strength line to smooth out the underlying trends. The period between the blue vertical lines reveals when the price trend was up but the relative trend was down.

There is not a lot we can do about this issue except be aware it exists. Trading strategies that rely on relative strength should always consider the trend of the benchmark as well.

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10.1 Advanced Applications of
Relative Strength

Basic relative strength analysis, as described above, can be used to judge relative trends in one-on-one comparisons. However, due to the nature of the calculation of the relative strength line, its values cannot be used to rank a universe based on relative strength. This is because the numerical values of these relative strength lines have no universal meaning. The output of the relative strength formula depends entirely on the prices of the securities being considered.

In Figure 10.1.2, we see two charts, each showing a weekly line chart and a relative strength line against the S&P 500 index. We have added a 30-period simple moving average to smooth the relative strength lines.

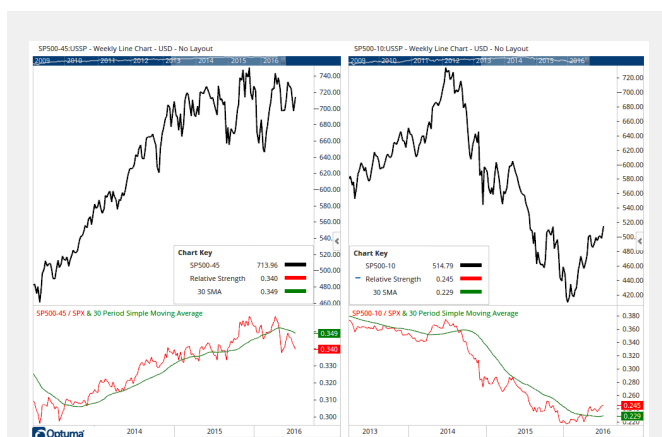


FIGURE 10.1.2 S&P 500 Information Technology (left) and S&P 500 Energy (right) with relative strength lines against the S&P 500 index

The chart on the left of Figure 10.1.2 shows the information technology sector with a relative strength value of 0.340 (MA at 0.349). The chart on the right is the energy sector with a relative strength value of 0.245 (MA at 0.229). However, the relative strength line of the energy sector has just started to move higher (and is above its moving average) while the relative strength of information technology has just begun to move lower (and is below its moving average).

Based on this observation, information technology is in a relative downtrend, while energy is in a relative uptrend. For a long-only portfolio, energy would be a better candidate. However, we cannot use their relative values alone (0.340 and 0.245) to rank these securities.

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10.1 Advanced Applications of Relative Strength

Introducing the Relative Rotation Graph

Relative Rotation Graphs®, or RRGs®, were created by Julius de Kempenaer for use in his sell-side research for institutional clients in 2002. De Kempenaer's philosophy was as follows:

In an over-brokered market where every portfolio manager and buy-side analyst receives hundreds of emails, reams of printed (research) reports, and dozens of phone calls every day, analysts need to give them something unique, something that adds value for them and something they do not get anywhere else.¹

As a sell-side analyst, it was common for de Kempenaer to visit clients to talk about their portfolios. Clients would often ask him which securities they should add to their portfolio, but unless de Kempenaer had undertaken a full analysis of all the stocks in the client's universe, there was no way to know.

RRGs were the solution to this problem. The RRG is a modified scatter plot that displays, in a single chart, all the securities that could be considered by a portfolio manager relative to a benchmark. It was then a simple matter to classify the securities and discern where the security's relative strength was likely to move in the future.

The building blocks for the RRG calculations are:

- JdK RS-Ratio (JdK Relative Strength-Ratio)
- Jdk RS-Momentum (JdK Relative Strength-Momentum)

JdK Relative Strength-Ratio

The JdK RS-Ratio is a normalized measure of relative trend that can be used across all the securities in a universe against the benchmark for that universe. It also allows for comparisons of the relative strength of one security in the universe against one or more other securities. The JdK RS-Ratio calculation measures the trend of the relative strength itself.

The value of the JdK RS-Ratio is centered at 100:

- If a security has a JdK RS-Ratio value of 100, it implies the security is moving in perfect harmony with the benchmark.
- Values greater than 100 imply the security is outperforming the benchmark.
- Values below 100 indicate that the security is underperforming the benchmark.

Having a numerical value for the JdK RS-Ratio allows an analyst to rank and compare all the securities in a universe without

Introducing the Relative Rotation Graph

SP500-55	S&P 500 - Utilities - Sector	356.54	1.05%	103.20
SP500-30	S&P 500 - Consumer Staples - Sector	816.85	-0.27%	101.50
SP500-25	S&P 500 - Consumer Discretionary - Sector	1,454.59	-0.41%	100.39
SP500-40	S&P 500 - Financials - Sector	687.41	0.40%	100.01
SP500-20	S&P 500 - Industrials - Sector	1,052.27	0.03%	99.98
SP500-35	S&P 500 - Health Care - Sector	1,655.45	-0.33%	99.84
SP500-45	S&P 500 - Information Technology - Sector	3,778.98	0.23%	99.74
SP500-50	S&P 500 - Communication Services - Sector	291.65	-0.17%	99.72
SP500-15	S&P 500 - Materials - Sector	571.31	-0.44%	99.43
SP500-60	S&P 500 - Real Estate - Sector	232.36	-0.90%	98.92
SP500-10	S&P 500 - Energy - Sector	710.73	-0.12%	98.90

FIGURE 10.1.3 S&P 500 Sectors Ranked by JdK RS-Ratio (as of May 8, 2024)

Figure 10.1.3 shows an example of the S&P 500 sectors universe ranked by the value of the JdK RS-Ratio indicator.

Plotting the JdK RS-Ratio below the price chart indicates the trend of the relative strength of the security being analyzed against its benchmark, which we can see in Figure 10.1.4. The visual inspection of the JdK RS-Ratio line is as straightforward as the interpretation of the relative strength line — a rising JdK RS-Ratio indicates that the security is in a relative uptrend and a declining JdK RS-Ratio implies the security is in a relative downtrend.



FIGURE 10.1.4 S&P 500 industrials with relative strength line and JdK RS-Ratio line

The JdK RS-Ratio is a uniform measure of relative trend that allows us to identify high values as strong and low values as weak. Plotting the indicator on a regular price chart indicates the direction and strength of the relative trend for that security.

When looking at this chart, it is easy to determine the direction of the JdK RS-Ratio line. However, what if we only want to look at a list of tickers? The final value of the JdK RS-Ratio does not tell us in which direction it is moving.

The issue can be seen in Figure 10.1.4. Values roughly at or above 104 can be considered “high values.” On December 9, 2022, the JdK RS-Ratio value passed through 104 while moving higher. A couple of months later, on February 10, 2023, the JdK RS-Ratio line again passed through 104, but now the line was moving lower.

Introducing the Relative Rotation Graph

RS-Ratio, we also need to consider the direction in which the JdK RS-Ratio line is moving. We can do this by considering momentum.

JdK Relative Strength-Momentum

To determine whether the JdK RS-Ratio line is moving up or down and at what pace, the JdK RS-Momentum line is used. The JdK RS-Momentum line uses a rate of change approach to gauge the direction and the pace of the relative trend, as measured by the JdK RS-Ratio line. The raw JdK RS-Momentum values are normalized in the same way as the JdK RS-Ratio values.

By plotting these two uniform measures on one chart, shown in Figure 10.1.5, the interaction between JdK RS-Ratio and Momentum becomes clear.

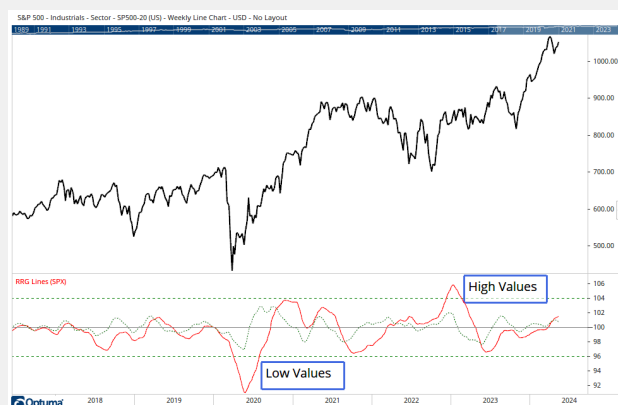


FIGURE 10.1.5 S&P 500 Industrial sector line chart top panel, and JdK RS-Ratio (red) and JdK RS-Momentum (green) bottom panel

JdK RS-Ratio is considered to be the most important and most influential numerical measure in the relative strength analysis. However, notice that the JdK RS-Momentum is leading the JdK RS-Ratio and serves as an early warning signal of a pending change in the JdK RS-Ratio.

When the JdK RS-Momentum drops below 100, the JdK RS-Ratio forms a peak and starts to move lower. When JdK RS-Momentum crosses above 100, the JdK RS-Ratio line forms a trough and starts to move higher.

Armed with this information, we can further interpret the JdK RS-Ratio values we looked at previously. In Figure 10.1.6, we have updated Figure 10.1.3 to include the JdK RS-Momentum values.

Ticker	Name	Last	Change (%)	JdK RS-Ratio	JdK RS-Momentum
SP500-55	S&P 500 - Utilities - Sector	356.54	1.05%	103.20	99.88
SP500-30	S&P 500 - Consumer Staples - Sector	816.85	-0.27%	101.50	99.61
SP500-25	S&P 500 - Consumer Discretionary - Sector	1,454.59	-0.41%	100.39	100.53
SP500-40	S&P 500 - Financials - Sector	687.41	0.40%	100.01	99.44
SP500-20	S&P 500 - Industrials - Sector	1,052.27	0.03%	99.98	99.64
SP500-35	S&P 500 - Health Care - Sector	1,655.45	-0.33%	99.84	99.85
SP500-45	S&P 500 - Information Technology - Sector	3,778.98	0.23%	99.74	100.83
SP500-50	S&P 500 - Communication Services - Sector	291.65	-0.17%	99.72	99.28
SP500-15	S&P 500 - Materials - Sector	571.31	-0.44%	99.43	99.77
SP500-60	S&P 500 - Real Estate - Sector	232.36	-0.90%	98.92	100.43
SP500-10	S&P 500 - Energy - Sector	710.73	-0.12%	98.90	97.92

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100, we can see only Consumer Discretionary also had positive JdK RS-Momentum, that is, a value greater than 100. This is a sector that would warrant further investigation.

Relative Strength

Comparing Large Universes

Using the RRG indicators in combination with price charts as described above is a structured approach to analyzing the trend in relative strength of the securities in a universe. This is a significant improvement over charts with only a relative strength line. However, even with this structured and comparative approach, it is still necessary to browse through all the individual securities of a universe to look at the charts. This may be feasible when the number of securities is small, but when we cover a large number of securities, there needs to be a better way.

¹ de Kempnaer, Julius. "Everything is Relative Strength is Everything." In *New Frontiers in Technical Analysis: Effective Tools and Strategies for Trading and Investing*, by Paul Ciana. New Jersey: Bloomberg Press, 2011.

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